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DEVICES FOR GENERATING VOTING RECEIPTS

Abstract

A device for generating a voting receipt is provided for use in a voting system. The voting system also includes an electronic voting machine configured to record the voting choices of a voter. The device communicates with the electronic voting machine via a local area network only, and receives from the electronic voting machine data indicating the voting choices of the voter. The device prints a voting receipt that indicates the voting choices of the voter. The voting receipt is a permanent record by which the voter can verify that their votes have been accurately recorded by the electronic voting machine.

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Background/Summary

BACKGROUND

[0001] The integrity of public elections has been, and continues to be a controversial topic in the

United States and elsewhere. The confidence of voters that their votes have been accurately recorded is critical to functioning democracies where the choice of political leaders and important public issues are resolved through voting by the citizenry. The issue of voter confidence can be especially problematic in jurisdictions that use electronic voting machines in which the voter's choices are converted into an electronic format that the voter cannot review to verify that voter's choices have been registered accurately.

SUMMARY

[0002] In one aspect of the disclosed technology, a device for generating a voting receipt is configured for use with a voting system having an electronic voting machine configured to record voting choices of a voter. The device includes a control unit configured to be communicatively coupled to the electronic voting machine via a local area network only, and to receive from the electronic voting machine data indicating the voting choices of the voter. The device also includes a printing unit communicatively coupled to the control unit and configured to print the voting receipt. The voting receipt indicates the voting choices of the voter.

[0003] In another aspect of the disclosed technology, the voting receipt further includes a time stamp indicating the time and date on which the voting choices of the voter were recorded by the electronic voting machine.

[0004] In another aspect of the disclosed technology, the control unit is configured to be communicatively coupled to the electronic voting machine via an encrypted communication link.

[0005] In another aspect of the disclosed technology, the control unit includes a memory, and the control unit is further configured to store in the memory the data indicating the voting choices of the voter.

[0006] In another aspect of the disclosed technology, the control unit is further configured to download the data indicating the voting choices of the voter to a device configured to receive the data indicating the voting choices of the voter.

[0007] In another aspect of the disclosed technology, the control unit is further configured to communicate only with the electronic voting machine and the device configured to receive the data indicating the voting choices of the voter.

[0008] In another aspect of the disclosed technology, the control unit is further configured to receive inputs only from the electronic voting machine and the device configured to receive the data indicating the voting choices of the voter.

[0009] In another aspect of the disclosed technology, the local area network includes the voting-device, the electronic voting machine, and the device configured to receive the data indicating the voting choices of the voter.

[0010] In another aspect of the disclosed technology, the control unit is further configured so that a running time and date upon which the time stamp is based cannot be altered by an outside source.

[0011] In another aspect of the disclosed technology, the control unit is further configured so that the data indicating the voting choices of the voter cannot be altered or deleted from the memory by an outside source.

[0012] In another aspect of the disclosed technology, the control unit includes the memory, a processor communicatively coupled to the memory, and computer-executable instructions stored on the memory and configured to, upon execution by the processor, cause the printing unit to generate the voting receipt.

[0013] In another aspect of the disclosed technology, the computer-executable instructions are open-source software.

[0014] In another aspect of the disclosed technology, a voting system includes an electronic voting machine configured to record voting choices of a voter, and a device for generating a voting receipt. The device includes a control unit communicatively coupled to the electronic voting machine via a local area network only, and configured to receive from the electronic voting machine data indicating the voting choices of the voter. The device also includes a printing unit

communicatively coupled to the control unit and configured to generate the voting receipt. The voting receipt indicates the voting choices of the voter.

[0015] In another aspect of the disclosed technology, the electronic voting machine is one of an optical scanner and a direct recording electronic machine.

[0016] In another aspect of the disclosed technology, the electronic voting machine and the device for generating voting receipts are provided by different manufacturers.

[0017] In another aspect of the disclosed technology, the electronic voting machine is configured to generate an output indicting the voting choices of the voter, and the device for generating a voting receipt is configured to generate the voting receipt upon receiving the output of the electronic voting machine.

[0018] In another aspect of the disclosed technology, the printout further incudes a time stamp indicating the time and date on which the voting choices of the voter were recorded by the electronic voting machine.

[0019] In another aspect of the disclosed technology, the device for generating a voting receipt is communicatively coupled to the electronic voting machine via an encrypted communication link.

[0020] In another aspect of the disclosed technology, the printing unit is configured to generate the voting receipt on a real-time or near real-time basis with the recordation of the voting choices of the voter by the electronic voting machine.

[0021] In another aspect of the disclosed technology, the control unit is further configured to download the data indicating the voting choices of the voter to a device configured to receive the data indicating the voting choices of the voter, and the control unit is further configured to communicate only with the electronic voting machine and the device configured to receive the data indicating the voting choices of the voter.

[0022] In another aspect of the disclosed technology, a process for verifying voting totals includes providing an electronic voting machine configured to record voting choices of a voter, and providing a device configured to generate a receipt indicating the voting choices of the voter as cast on the electronic voting machine, and further configured to store data indicating the voting choices of the voter as cast on the voting machine, the device being physically and functionally independent of the electronic voting machine. The process also includes providing the receipt to the user, and publishing the data indicating the voting choices of the voter as stored in the device in a medium accessible to the voter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The following drawings are illustrative of particular embodiments of the present disclosure and therefore do not limit the scope of the present disclosure. Embodiments of the present disclosure will hereinafter be described in conjunction with the appended drawings, wherein like numerals denote like elements.

[0024] FIG. 1 is a diagrammatic illustration of a voting system that includes a device for generating voting receipts;

[0025] FIG. 2 is a schematic illustration of the receipt-generating device shown in FIG. 1; and

[0026] FIG. 3 is a schematic illustration of a control unit of the receipt-generating device shown in FIGS. 1 and 2.

DETAILED DESCRIPTION

[0027] The inventive concepts are described with reference to the attached figures, wherein like reference numerals represent like parts and assemblies throughout the several views. The figures are not drawn to scale and are provided merely to illustrate the instant inventive concepts. The figures do not limit the scope of the present disclosure or the appended claims. Several aspects of

the inventive concepts are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide a full understanding of the inventive concepts. One having ordinary skill in the relevant art, however, will readily recognize that the inventive concepts can be practiced without one or more of the specific details or with other methods. In other instances, well-known structures or operation are not shown in detail to avoid obscuring the inventive concepts.

[0028] The figures depict a device **10** for generating voting receipts. The device **10** is configured for use with a voting system **12**. The voting system **12** comprises a plurality of electronic voting machines **20** by which voters can cast their votes for a particular candidate for elective office, or a particular issue on the ballot. Each electronic voting machine **10** is communicatively coupled to a respective device **10**. The device **10** generates a printed record in the form of a receipt **30** that displays the voting choices of the voter, i.e., the names of the candidates for which the voter has cast a vote, and/or the voter's choices on any issues on the ballot. Only one of the devices **10** and one of the voting machines **20** are depicted in FIG. **1**, for clarity of illustration.

[0029] The electronic voting machine **20** can be, for example, a direct recording electronic machine as depicted in FIG. **1**. The electronic voting machine **20** is configured, for example, with a touchscreen display **21** that shows the names of the candidates for elective office and/or the public issues to be voted upon. The voters can cast their votes by pressing the appropriate virtual button on the display **21**. Alternative embodiments of the system **10** can be equipped with direct recording electronic machines having other configurations.

[0030] In other alternative embodiments, the electronic voting machine **20** can be an optical scanner, or optical scan tabulator, configured to scan a ballot that has been filled out or completed by the voter. The completed ballot indicates the voter's choice of a particular candidate for elective office, and/or the voter's choice in a particular issue listed the ballot. The optical scan tabulator can include a scanning device, and a tabulator communicatively coupled to the scanning device. The tabulator is configured to tabulate the votes for a particular candidate or issue as indicated on the completed ballot, based on the scan of the completed ballot generated by the scanning device. The ballots can be hand-marked by the voter. Alternatively, the marked ballots can be generated by an electronic device, such as a ballot marking device, in response to the voter's input thereto.

[0031] The device **10** can be located adjacent to the voting machine **20**, so that the voter can retrieve the receipt **30** immediately upon its generation. In applications where the voting machine **20** is enclosed by a screen or other privacy provision, the device **10** can be located within the screened-off area to help ensure that the privacy of the voter is maintained. Typically, one device **10** is associated with one voting machine **20**, to help ensure that the actual receipt **30** provided to the voter accurately reflects that particular voter's choices.

[0032] Referring to FIG., **2**, the device **10** comprises a printing unit **32**, a control unit **34** communicatively coupled to the printing device **32**, and a display **35** communicatively coupled to the control unit **34**. The printing device **32** can be configured as a conventional laser printer comprising, for example, a laser, a fusing unit, a toner and toner cartridge, a drum unit, a transfer belt, and other components that are arranged, and function in a conventional manner. The printing device **32** can be configured as another type of printer, such as a dot matrix printer, an inkjet printer, an LED printer, etc., in alternative embodiments of the device **10**.

[0033] Referring to FIG. **3**, the control unit **34** comprises a processor **36**, such as a microprocessor; an internal bus **38**; a memory **40** communicatively coupled to the processor **36** via the bus; computer-executable instructions **42** stored in the memory **40**; and an input-output interface **44** communicatively coupled to the internal bus **38**. The control unit **34** can have other configurations in alternative embodiments. Also, the control unit **34** can include additional components, a description of which is not necessary to an understanding of the technology disclosed herein.

[0034] The display **35** can be configured to provide an indication of the status of the device **10**. For example, the display **35** may display whether the device **10** is on-line and properly communicating

with its associated voting machine **20**.

[0035] Unlike voting machine software, the program logic embedded in the computer-executable instructions **42** can be open source, so that the program logic can be read by anyone. Thus, the program logic is proprietary, but is not hidden or restricted from examination by others.

[0036] In addition to the voting choices of the voter, the receipt **30** also can include a time stamp indicating the time and date on which the receipt **30** was generated. Other information can be included on the receipt **30** in alternative embodiments. For example, the receipt **30** can display the voting location, an identifier of the particular voting machine **20** to which the device **10** is connected, etc.

[0037] The time stamp is generated by the device **10** independent of the voting machine **20**. The time and date upon which the time stamp is based can be set when the control unit **34** of the device **10** is manufactured, and the control unit **34** can be configured so that the running time and date cannot be altered subsequently by an outside source, i.e., by a source other than the manufacturer of the device **10**. These features can help to reduce, or virtually eliminate the possibility for the time stamp to be manipulated by a malevolent actor attempting to cause a fraudulent time stamp to be placed on the receipt **30**.

[0038] The receipt **30** is generated on a real-time or near real-time basis with the casting of the vote, so that the receipt **30** is available to the voter immediately upon completion of the vote-casting process. In alternative embodiments, the device **10** can be configured to receive an input from the voter regarding whether the voter wishes for the receipt **30** to be printed. The voter can be prompted for the input, for example, by a message displayed on the display **35**. The user can provide the input, for example, using a virtual button on the display **35** in applications where the display **35** is a touchscreen display. This feature can help to enhance voter privacy by avoiding the printing of a receipt **30** that may be generated without the voter's knowledge and inadvertently left in the device **10** after the voter has left the voting booth.

[0039] In addition to generating the receipt **30**, the device **10** can be configured to record, in the memory **40** of the control unit **34**, the votes cast by the voter exactly as the voting choices appear on the receipt **30**, helping to ensure that the information presented to the voter on the receipt **30** is a correct copy of the votes that were cast by the voter. The memory **40** can be configured so that the voting data, once recorded, cannot be deleted or altered by an outside source, i.e., by a source other than the manufacturer of the device **10**. The device **10** thus can maintain a separate and permanent record of the votes cast on the voting machine **20**. The control unit **34** can be configured to download the stored voting data periodically, to a device (not shown) specifically configured to be temporarily connected to the device **10** and to receive the download upon prompting the device **10**.

[0040] The device **10** is communicatively coupled to the voting machine **20** or other vote-casting machine by a suitable wired or wireless means. For example, FIG. **1** depicts a wired connection between the device **10** and the voting machine **20**. In alternative embodiments, the device **10** can be communicatively coupled to the voting machine **20** by, for example, a Wi-Fi connection. The wired or wireless communication link between the voting machine **20** and the device **10** can be encrypted, to help maintain voter privacy and to prevent hacking. The device **10** is not configured to be communicatively coupled or otherwise connected to any other components or any networks within, or outside of the voting system **12** (including the internet), other than the device configured to receive a download of the voting data stored in the control unit **34**. Thus, the device **10** is configured to communicate only over a local network comprising the voting machine **20** and the device specifically configured to receive the download of the voting data.

[0041] The device **10** is a separate component from the voting machine **20**. The device **10** can be manufactured by, and obtained from a source other than the manufacturer or provider of the voting machine **20**, to help ensure the integrity of the device **10** as an accurate and independent record of the votes cast by the voter.

[0042] The device **10** receives no direct input from the voter (other than a prompt to generate a

receipt **30**, in some possible embodiments). The device **10** is not used in any manner to cast or officially record votes, and does not require or alter any method of voting. The device **10** does not control the voting machine **20** in any way, does not provide any inputs to the voting machine **20**, and does not interact with the voting machine **20** other than receiving the data indicating the voter's selections, in a specific format recognizable by the device **10**. The device **10** does not recognize any other type of inputs, or any commands other than a command to download the voter data stored in the memory **40** of the control unit **34**. The device **10** cannot be accessed by the voting machine **20** or any other computing device by any means (other than the device to which the voting data is periodically downloaded).

[0043] The device **10** receives no input from the voting machine **20**, other than the line of data indicating the voter's selections. The device **10** receives no input from any other component within, or outside of the voting system **12** (other than the device to which the voting data is periodically downloaded). Therefore, there is no way to alter the program logic of the device **10**, or the voter data stored therein, and the device **10** is safe from any hacking or intentional manipulation of its data. The sole functions of the device **10** are to print the voter receipt **30** based on the voting selections relayed to the device **10** by the voting machine **20**, and to store a record of what was printed.

[0044] As noted above, the voting machine **20** is configured to provide an output to the device **10** indicating the votes cast by the voter. The voting machine **20** otherwise is unaffected by its interface with the device **10**. The device **10** can be used with any computerized electronic voting machine that provides the appropriate output. In jurisdictions where a state legislature or election authority requires the use of a device, such as the device **10**, that provides a voter receipt and/or a supplemental record of the voter's selections, the state legislature or the election authority can specify that the manufacturer of the voting machine **20** make the voting machine **20** compatible with the device **10**, i.e., the state legislature or election authority can specify that the voting machine **20** output a line of data, in the format recognizable by the device **10**, indicating the choices of the voter as recorded by the voting machine **20**.

[0045] The user can check the receipt **30** before leaving the voting area, to help ensure that the voter's selections have been recorded correctly by the voting machine **20**. In the event the user detects a discrepancy between the user's selections and the selections listed reflected on the receipt **30**, the user can notify an election worker before leaving the voting area, so that appropriate corrective action can be taken.

[0046] Also, because the device **10** records the voter's selections exactly as the selections appear on the receipt **30**, the voter can see, and confirm as accurate the voting data that has been stored in the device **10**. Thus, the voting totals documented by and stored in the devices **10** are backed up by the voting public. After an election, the official (legal) vote totals cast on electronic voting machines **20** should match the voting data stored in the corresponding devices **10**. If it does not, the public can verify that the votes totals stored in the devices **10** are the correct totals. The public can verify their votes, for example, on a website where the vote totals stored in the devices **10** used in that particular election are published. If significant discrepancies appear between the official vote count and the verified data from the devices **10**, an appropriate investigation can be undertaken by the election authority or other governmental actor. Thus, the use of the devices **10**, which are physically and functionally independent from the voting machines **20** and maintain an independent and non-alterable record of the voter's selections, can bolster public confidence in the integrity of the final vote count. By comparison, if the receipts **30** were printed by the voting machines **20**, the voting public would only have the word of the election authority and/or the voting machine provider that the voting selections on the receipts **30** match what has been tallied by the voting machines **20**.

[0047] The device **10** also can be used in connection with mail-in voting ballots. In such applications, the receipt **30** can be generated in the above-described manner when the mail-in ballot

is scanned by the voting machine **20**, and the receipt **30** can be mailed to the voter.

[0048] Although the present solution has been illustrated and described with respect to one or more implementations, equivalent alterations and modifications will occur to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In addition, while a particular feature of the present solution may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application. Thus, the breadth and scope of the present solution should not be limited by any of the above described embodiments. Rather, the scope of the present solution should be defined in accordance with the following claims and their equivalents.

Claims

1. A device for generating a voting receipt, the device being configured for use with a voting system comprising an electronic voting machine configured to record voting choices of a voter, the device comprising: a control unit configured to be communicatively coupled to the electronic voting machine via a local area network only, and to receive from the electronic voting machine data indicating the voting choices of the voter; and a printing unit communicatively coupled to the control unit and configured to print the voting receipt, wherein the voting receipt indicates the voting choices of the voter.
2. The device of claim 1, wherein the voting receipt further includes a time stamp indicating the time and date on which the voting choices of the voter were recorded by the electronic voting machine.
3. The device of claim 1, wherein the control unit is configured to be communicatively coupled to the electronic voting machine via an encrypted communication link.
4. The device of claim 1, wherein the control unit comprises a memory, and the control unit is further configured to store in the memory the data indicating the voting choices of the voter.
5. The device of claim 4, wherein the control unit is further configured to download the data indicating the voting choices of the voter to a device configured to receive the data indicating the voting choices of the voter.
6. The device of claim 5, wherein the control unit is further configured to communicate only with the electronic voting machine and the device configured to receive the data indicating the voting choices of the voter.
7. The device of claim 6, wherein the control unit is further configured to receive inputs only from the electronic voting machine and the device configured to receive the data indicating the voting choices of the voter.
8. The device of claim 5, wherein the local area network consists of the voting-device, the electronic voting machine, and the device configured to receive the data indicating the voting choices of the voter.
9. The device of claim 2, wherein the control unit is further configured so that a running time and date upon which the time stamp is based cannot be altered by an outside source.
10. The device of claim 4, wherein the control unit is further configured so that the data indicating the voting choices of the voter cannot be altered or deleted from the memory by an outside source.
11. The device of claim 4, wherein the control unit comprises the memory, a processor communicatively coupled to the memory, and computer-executable instructions stored on the memory and configured to, upon execution by the processor, cause the printing unit to generate the voting receipt.
12. The device of claim 11, wherein the computer-executable instructions are open-source software.
13. A voting system, comprising: an electronic voting machine configured to record voting choices

of a voter; and a device for generating a voting receipt, comprising: a control unit communicatively coupled to the electronic voting machine via a local area network only, and configured to receive from the electronic voting machine data indicating the voting choices of the voter; and a printing unit communicatively coupled to the control unit and configured to generate the voting receipt, wherein the voting receipt indicates the voting choices of the voter.

14. The system of claim 13, wherein the electronic voting machine is one of an optical scanner and a direct recording electronic machine.

15. The system of claim 13, wherein the electronic voting machine and the device for generating voting receipts are provided by different manufacturers.

16. The system of claim 13, wherein the electronic voting machine is configured to generate an output indicating the voting choices of the voter, and the device for generating a voting receipt is configured to generate the voting receipt upon receiving the output of the electronic voting machine.

17. The system of claim 13, wherein the printout further includes a time stamp indicating the time and date on which the voting choices of the voter were recorded by the electronic voting machine.

18. The system of claim 13, wherein the device for generating a voting receipt is communicatively coupled to the electronic voting machine via an encrypted communication link.

19. The system of claim 13, wherein the printing unit is configured to generate the voting receipt on a real-time or near real-time basis with the recordation of the voting choices of the voter by the electronic voting machine.

20. The system of claim 13, wherein: the control unit is further configured to download the data indicating the voting choices of the voter to a device configured to receive the data indicating the voting choices of the voter; and the control unit is further configured to communicate only with the electronic voting machine and the device configured to receive the data indicating the voting choices of the voter.

21. A process for verifying voting totals, comprising: providing an electronic voting machine configured to record voting choices of a voter; providing a device configured to generate a receipt indicating the voting choices of the voter as cast on the electronic voting machine, and further configured to store data indicating the voting choices of the voter as cast on the voting machine, the device being physically and functionally independent of the electronic voting machine; providing the receipt to the user; and publishing the data indicating the voting choices of the voter as stored in the device in a medium accessible to the voter.
